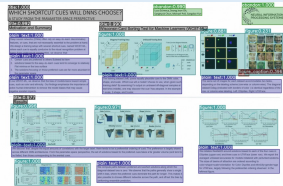


poster image

Stage 1: Layout Detection



Detected Elements



Image-like



Text-like

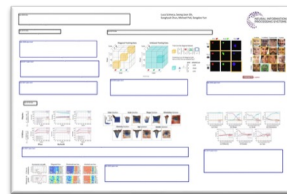


Decorative



Stage 2: Font Size Estimation

HTML scaffold



Stage 3: Design Captioning

Global Attributes

color palette



typography family



decorative motifs



Per-Region Design

spatial composition



layer interaction



typographic hierarchy



visual anchor



content density



emphasis strategy



color scheme



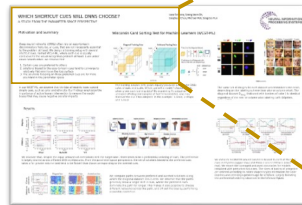
decoration treatment



Stage 4:

HTML Reconstruction

render



reconstructed poster

origin poster (zoom in)

WHICH SHORTCUT CUES WILL DNNs CHOOSE?

A STUDY FROM THE PARAMETER-SPACE PERSPECTIVE

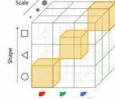
Motivation and Summary

Deep neural networks (DNNs) often rely on easy-to-learn discriminatory features, or cues, that are not necessarily essential to the problem at hand. We design a training setup with several shortcut cues, named WCST-ML, where each cue is equally conducive to the visual recognition problem at hand. Even under equal opportunities, we observe that:

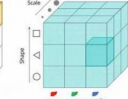
1. Certain cues are preferred to others
2. solutions biased to the easy-to-learn cues tend to converge to relatively flat minima on the loss surface
3. The solutions focusing on those preferred cues are far more abundant in the parameter space

Wisconsin Card Sorting Test for Machine Learning

Diagonal Training Data



Unbiased Testing Data



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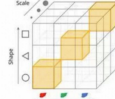
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